



Original article

Successional colonization of temporary streams: An experimental approach using aquatic insects



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ABSTRACT

The metacommunity concept studies the processes that structure communities on local and regional scales. This concept is useful to assess spatial variability. However, temporal patterns (e.g., ecological succession and colonization) are neglected in metacommunity studies, since such patterns require temporally extensive, and hard to execute studies. We used experimental habitats in temporary streams located within the Brazilian Cerrado to evaluate the importance of succession for the aquatic insect metacommunity. Five artificial habitats consisting of wrapped crushed rock were set transversally to the water flow in five streams. The habitats were sampled weekly to assess community composition, and replaced after sampling to identify new potential colonizers. We analyzed the accumulation of new colonizers after each week using a logistic model. We selected pairs of experimental habitats and estimated the Bray-Curtis dissimilarity index to assess the community composition trajectory during the experiment. We used the dissimilarity values in ANOVA tests, identifying the importance of time and space for the community. The number of new taxa stabilized in the third week, and we estimated a weekly increase of 1.61 new taxa in the community after stabilization. The overall pattern was a small change on community composition, but one stream had a higher weekly turnover. Our results showed a relevant influence of time in the initial communities of aquatic insects of temporary streams. However, we must observe the temporal pattern in a spatial context, once different streams have different successional history regarding number of taxa and community turnover. We highlight the importance of aerial dispersal and movement to seek oviposition sites as an important factor in determining colonization patterns.

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1. Introduction

The study of colonization processes and patterns has been gaining prominence within the field of community ecology in recent decades (Vellend, 2010). Historically, colonization has been reported as the main determinant of community succession, with species showing a tradeoff between traits that favor dispersal and persistence (Herben and Goldberg, 2014; Ricklefs, 1987). Species with high dispersal ability can colonize recently created or remote

habitats, causing species occupancy to become a dynamic process in biological communities (Hanski and Ovaskainen, 2000). Dispersal is important for community structure in ephemeral habitats, as it determines the initial species to settle. The ability of the species to migrate and colonize new sites is key for community organization in ephemeral habitats because the species in their communities hardly reach stability in their population dynamics (Altermatt and Ebert, 2010; Travis and Dytham, 1999).

Dispersal also leads to joint population dynamics and functionally connected communities across different sites. The relationship between communities linked by dispersal have been studied based on the metacommunity framework (Heino et al., 2015; Leibold et al., 2004). The main goal of the metacommunity theory is to explain the relative importance of dispersal and the

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interaction between the organisms and the environment in creating and maintaining communities (Winegardner et al., 2012). Dispersal among communities tends to increase the number of species (or higher taxa) and interactions (e.g., competition, predation and habitat selection), and to reduce local diversity (Caley and Schluter, 1997; He et al., 2005). When ephemeral habitats are considered in the context of the metacommunity theory, the balance between a fast colonization and competition for limited space would drive local diversity. Therefore, ecological succession in ephemeral habitats would be highly dependent on dispersal among communities, in contrast with a more stable environment, where successional patterns are largely related with local biological interactions (Murrell et al., 2014).

Despite the metacommunity concept has advanced our understanding of how spatial dynamics and local interactions structure communities, such advances are recent and more empirical evidence is required (Logue et al., 2011; Winegardner et al., 2012). Temporary streams can be used as controlled sites to assess successional patterns arising from early colonization. The aquatic insects inhabiting temporary streams are a good metacommunity model to use in successional studies, provided that transient habitat availability selects for the occurrence of species with faster dispersal across the spatial range of the metacommunity and those that develop within the timeframe when water is available. These unique attributes (ephemeral habitat availability and dispersal dependence) make temporary streams and the associated aquatic insect communities good tools to test the metacommunity concept, particularly as regards to successional patterns and processes.

The succession of insects in freshwater assemblages is affected by the age of the habitat and by other extrinsic factors such colonization and season (Ruhí et al., 2012). However, the relative contribution of extrinsic effects versus age-dependence effects on species variation is unclear, with few studies about this theme. If age-dependence is the most important factor, the metacommunity will show a largest species turnover along the temporal continuum. On the other hand, if extrinsic factors are more relevant, the highest species turnover will be observed between sites and the decay of community similarity in a spatial gradient. Thus we elaborated five hypotheses regarding the aquatic insect community dynamics in temporary streams: (1) the number of new taxa accumulated in the local community will decrease over time after colonization begins; (2) taxa turnover is a result of community aging; (3) the established communities will be different from the communities observed in recently colonized habitats; (4) community turnover will be constant with the passage of time; (5) the spatial component is important for community structure because stream colonization depends on dispersal.

The objective of this study was to test the five hypotheses above. We also evaluated the effect of time on community structure at each stream, and predicted a decrease community similarity over time. We also predicted the occurrence of different successional history for each local community due to differences in their pools of colonizers.

We evaluated our hypotheses in five streams in the Cerrado biome, which dry naturally once a year. We used an innovative study design that applies elements of the state-space concept. A state-space design depicts if the dynamics of a variable of interest (in our study the composition of the aquatic insect community) are related to their previous state (Pedersen et al., 2011). Other elements may be added to the design to discern other important factors, such as environment and space. We designed a study in which we were able to isolate each phase of community development in temporary streams, during a period in which the aquatic insect communities were expected to stabilize (Landeiro et al., 2010).

2. Material and methods

2.1. Study area and experiment design

The five temporary streams are located near Niquelandia, GO, Brazil (14°10'24"S, 48°21'47"W). The area has a semi-humid tropical climate, with rainy season from October to April, and a dry season from May to September. Small streams in this region dry during the dry season, and flood once again in the mid rainy season. Despite these small streams flood intermittently, there are two large, permanent rivers in the region, Maranhão and Das Almas that provide a pool of stream colonizers. The nearest permanent river to the studied streams is the Das Almas River (1.4–4 km from the experimental sites). Annual water temperatures are on average 24 °C (daily values ranged 23–27 °C; INMET, 2015) with pH ranging from 7.2 to 8.9. The closed tree canopy in the riparian zone prevents much of the sunlight from reaching the ground and water surface.

A total of five artificial habitats were placed transversal to the water flow and at least one meter from each other in each stream (25 experimental units) in the early rainy season of 2008. The artificial habitat consisted of nylon net bags filled with one kilogram of crushed rock (5 mm mesh size). We have chosen this kind of experimental unit because submerged rock is a representative habitat in Cerrado streams (Bispo and Oliveira, 2007). The sampling begun one week after the deployment of the artificial substrates. We collected one experimental unit per stream per week. The experiment lasted five weeks (35 days). At the time of collection, experimental units were stored in a plastic bottle with alcohol to prevent the loss of specimens.

The experimental units collected each week were replaced by new experimental units deployed at the same location. These new experimental units remained in the stream for only one week and were removed in the next sampling date (Fig. 1a). This procedure was carried out to detect variation in potential colonizers at different times as it allows a new colonizer to occupy the habitat without a pre-established community, relaxing the influence of biological interactions. Therefore, we used 45 experimental units in the study, 25 placed at the beginning of the experiment and 20 deployed to replace the sampling units collected each week. We counted and identified specimens of the orders Ephemeroptera and Diptera to the family level, because many of these specimens are fragile and became damaged after collection. Specimens in the orders Plecoptera, Trichoptera, Odonata, Heteroptera, Megaloptera and Coleoptera were identified to the genus level.

2.2. Statistical analyses

We tested the five hypotheses using two different procedures, because the hypotheses were related to different community parameters. For the first hypothesis we adjusted a logistic function for taxon accumulation in each week. This model had the advantage of identifying informative parameters in the succession event, such as the initial and established number of taxa in the community. The model is described by the following equation:

$$S_w = E / \left[1 + (E - I) / I \times e^{-a \times w} \right],$$

where S_w is the number of taxa in week w , E is the number of taxa established in the end of succession, I is the number of taxa in the beginning of the experiment and a is the rate of changes in number of taxa through time. After estimating the parameter, we used the F ratio to validate the model. We also compared the number of taxa in each week and stream by means of a two-way ANOVA. In these analyses we considered only the experimental units placed at the

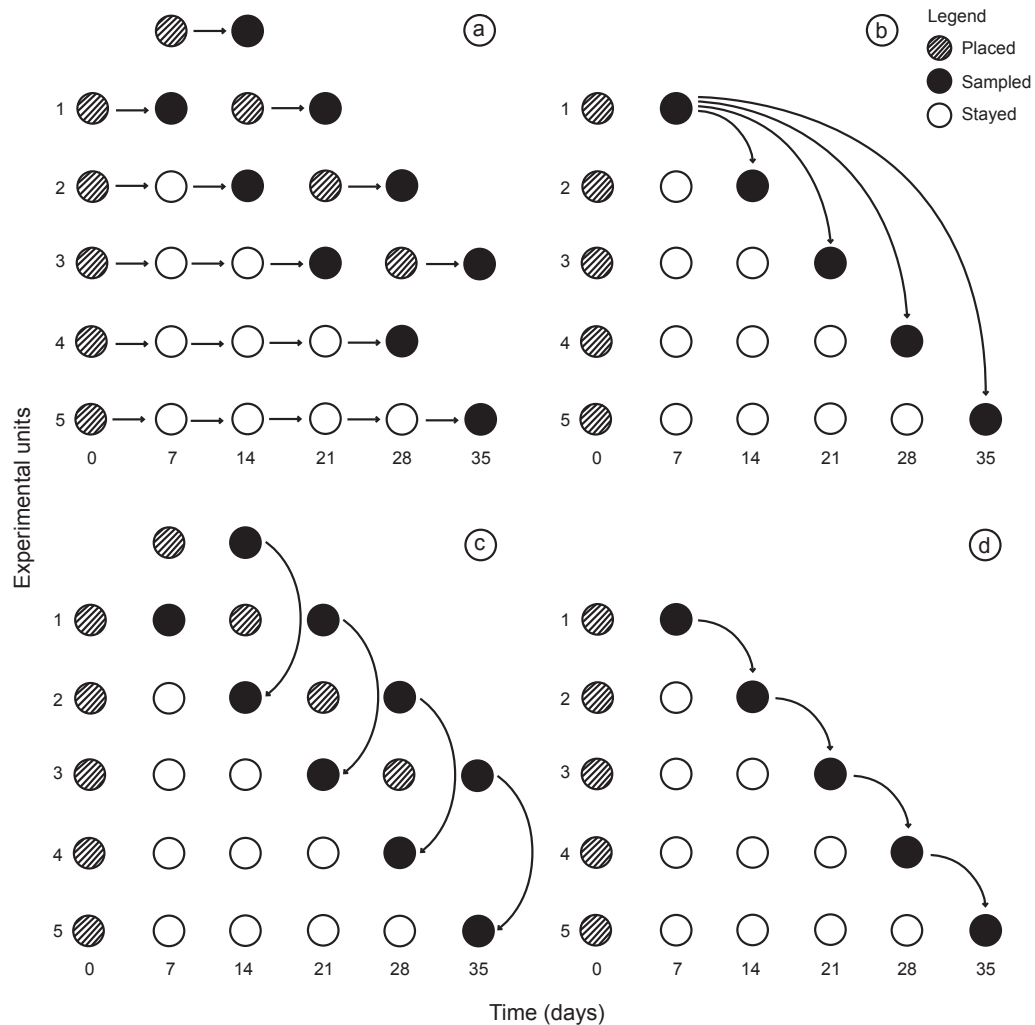


Fig. 1. Diagram of the experimental design and hypotheses being tested in regards to the influence of time in the structure of communities in temporary streams. (a) The experimental design shows the artificial habitats used, where the arrows represent the same experimental unit (Y-axis) in the course of time (X-axis). (b), (c) and (d) The experimental units selected in pairs to be used to test the hypothesis II, III and IV respectively, where the arrows represent the pairs used. In every diagram, the dashed circle is the experimental unit placed on a given moment, the black circle is the experimental unit sampled, and the white circle is the artificial habitat which remained in the experiment.

beginning of the experiment.

We tested hypotheses 2 to 4 using pairwise community dissimilarity (Bray-Curtis index) between selected experimental units. The selection of the pairs of experimental units depended on the hypothesis tested. We used week as discrete temporal variable to assess differences in community maturation. The second hypothesis evaluates community shifts in the course of time and predicts that differences from the initial community increase as time passes. We used the community dissimilarity between the experimental units sampled in the first week and other four samples distributed among the four successive weeks to test this hypothesis (Fig. 1b).

To test the difference between old and new communities, we compared the experimental units placed at the beginning of the experiment with those placed on the previous week within the same stream (third hypothesis, Fig. 1c). We have also determined the space-state community turnover (Pedersen et al., 2011) using the weekly experimental units sampled successively, in pairs (fourth hypothesis, see Fig. 1d). In all three cases (hypotheses two to four), we had four experimental replicates in each stream (Fig. 1), that is, 20 samples to test. We evaluated the second, third and fourth hypothesis using the ANOVA test, specifically a two-way

ANOVA with the week as fixed factor and the stream as a random factor. However, we do not consider the interaction between time (week) and site (stream), since we restricted the hypothesis to the general succession pattern despite the local colonization sequence. All analyses were carried out using the statistical software R (R Development Core Team, 2014).

3. Results

We sampled 16,935 individuals, comprising 32 operational taxonomic units (OTUs, nine families of Ephemeroptera and Diptera, and 23 genera of Plecoptera, Trichoptera, Odonata, Heteroptera, Megaloptera and Coleoptera). We recorded on average $5.49 (\pm 3.18)$ OTUs per experimental unit, with the genus *Smicridea* as the most abundant OTU (11,662 individuals), followed by Chironomidae (2204 individuals), Simuliidae (1370) and Baetidae (988). The abundance of the other OTUs ranged from 1 to 430 individuals (27.33 individuals in average). Chironomidae occurred in 86% of the experimental units and was the OTUs with the highest prevalence.

We recorded a large number of taxa in the first week of the experiment (15 OTUs, 46% of the observed taxa), especially in the first week (Fig. 2). Odonata (four genera), Diptera and

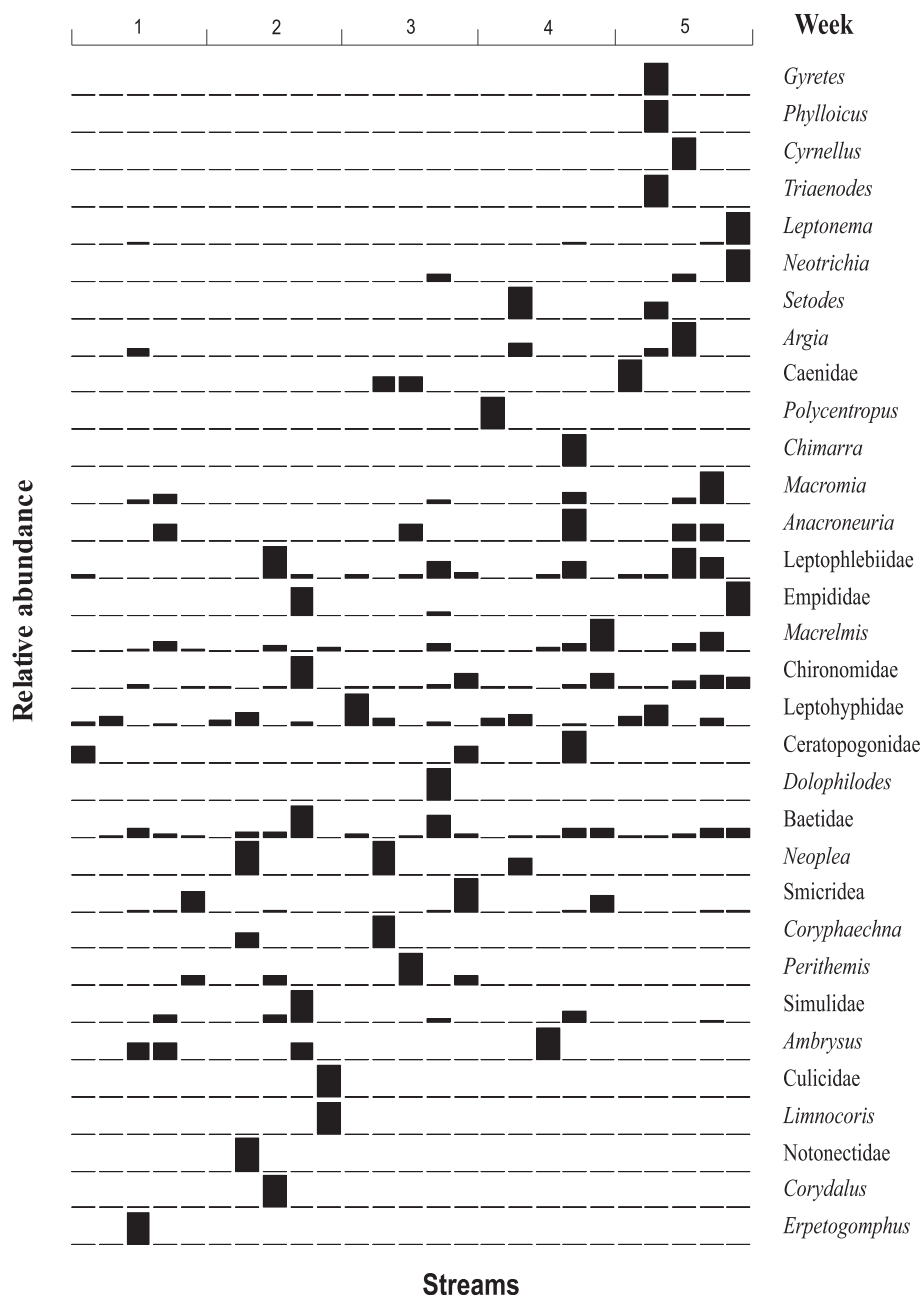


Fig. 2. Relative abundance of taxa in different weeks at the five streams.

Ephemeroptera (three families each) represented the initial community. The communities occurring in the third and fourth weeks had 11 different taxa, where the order Heteroptera was numerically dominant in community composition (three genera). The number of taxa that occurred only in the last week was reduced (six genera), and consisted mostly of Trichoptera (five genera). Baetidae, Chironomidae and Leptohyphidae occurred every week.

The number of taxa exclusive to habitats placed after the first week was small (only one family and one genus). A larger number of taxa were exclusively found in the experimental units placed at the beginning of the experiment (12 OTUs). The number of taxa that occurred in both types of experimental units represented most of the observed taxa (18 OTUs). The number of new OTUs per week stabilized in the third week after the beginning of the experiment (Fig. 3). We estimated an average increase of 1.61 new taxa per

week after community stabilization, and average colonization of 6.4 OTUs at the beginning of experiment. The logistic function explained a great portion of the variation in taxon accumulation ($F_{2,20} = 13.88$, $P < 0.01$ and $r^2 = 0.57$). The model used was appropriate, since the lack-of-fit test showed a small variance surrounding the curve ($F_{2,20} = 1.95$, $P = 0.73$). The result supports hypothesis one.

The number of taxa differed among streams, ranging from three to 10 taxa per stream ($F_{4,16} = 9.52$, $P < 0.01$ and $R^2 = 0.63$, Fig. 4). The mean number of taxa did not differ between weeks ($F_{4,16} = 1.69$, $P = 0.20$).

Three results were important for the successional process: (1) community aging had no influence on taxa turnover (Table 1, first ANOVA result), (2) the structure of established communities was equivalent to that of recently colonized habitats (Table 1, second

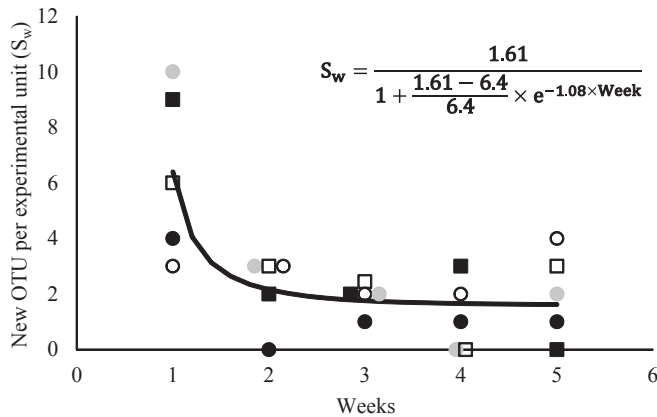


Fig. 3. Weekly accumulation of taxa in each stream. Individual streams are represented by different symbols.

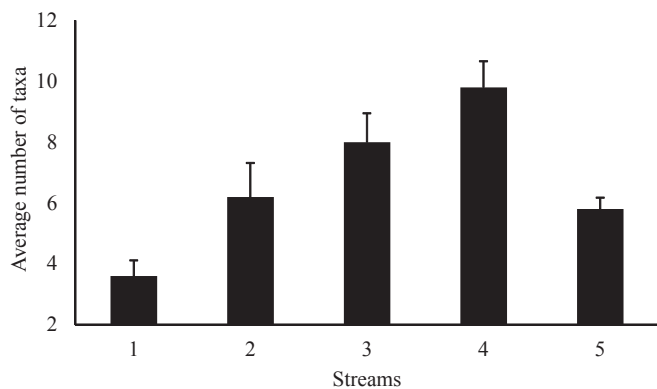


Fig. 4. Average number of taxa in each temporary stream.

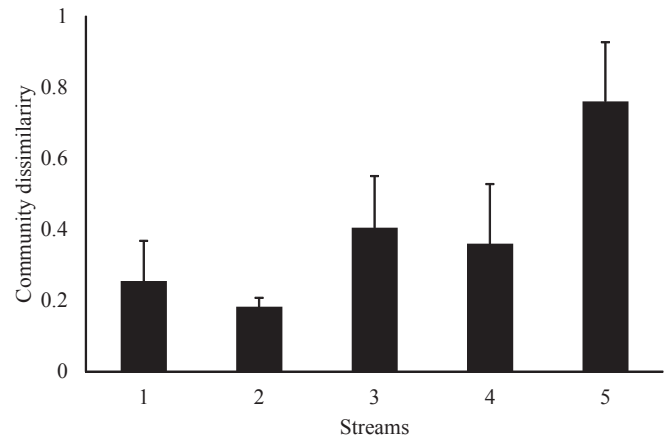


Fig. 5. Dissimilarity in community composition (Bray-Curtis dissimilarity) when observing taxa variation in one-week intervals for each stream.

4. Discussion

The metacommunity theory is a key concept in explaining community patterns in space and time, but the metacommunity concept has a void regarding successional processes and how time influences individual communities (Munguia, 2014). Our results showed a relevant influence of time on the initial aquatic insect communities of temporary streams, and corroborate the first and fifth hypotheses of our study (reduction in the accumulation of new taxa over time, after colonization begins; and the importance of the spatial component for community structure). However, we must observe the temporal pattern in a spatial context, because different streams have different successional histories regarding number of taxa and community turnover. These results show the importance

Table 1

Effects of different temporal and spatial elements on the turnover of aquatic insects in temporary streams.

Hypotheses	Source of variation	D.F.	S.S.	M.S.	F	r ²	P
Turnover and community age (II)	Time	3	0.08	0.03	0.32	0.07	0.81
	Stream	4	0.15	0.04	0.46	0.12	0.77
	Residual	12	0.99	0.08			
New vs. established communities (III)	Time	3	0.37	0.11	1.50	0.26	0.26
	Stream	4	0.07	0.02	0.20	0.05	0.93
	Residual	12	0.98	0.08			
Constant changes in community (IV)	Time	3	0.38	0.13	2.11	0.20	0.15
	Stream	4	0.80	0.20	3.37	0.42	0.04
	Residual	12	0.71	0.06			

Differences ($p < 0.05$) are shown in boldface.

ANOVA result), and (3) taxa turnover was constant along the experiment, regardless of the comparison made (Table 1, third ANOVA result). These results refute the second and third hypothesis, but moderately validate to fourth hypothesis. Overall, the replaced experimental units shared half of their community with the units placed at the beginning of the experiment (Bray-Curtis dissimilarity = 0.45 ± 0.27), and a similar proportion of taxa was shared among the initial and final communities (Bray-Curtis dissimilarity = 0.43 ± 0.23). Space had a significant effect on the weekly community changes (Table 1, small changes in community). We observed moderate weekly changes in community composition (Fig. 5; $39 \pm 22\%$ weekly taxa replacement). However, stream five showed a higher turnover ($76 \pm 16\%$ weekly taxa replacement). This result corroborates the hypotheses five.

of successional dynamics in temporary habitats. We emphasize the experimental basis of this study, which provides relevant results concerning the turnover of aquatic insect community in temporary streams and adds empirical evidence to the metacommunity theory.

We must, however, consider that the identification of all orders at same taxonomic levels was unfeasible due to the conservation status of the specimens. Therefore, our conclusions need to be taken with caution. The use of different taxonomic resolution levels could be a source of error, since many genera could be aggregate in only one family, leading to a lower number of OTUs in an experimental unit. However, the abundance of taxa identified to the family level, was restricted to few units in the same week or distributed among all experimental units when all weeks were

considered. Therefore, identification at different taxonomic resolution levels had probably little influence on the results on turnover of OTUs, since the probability of different species of a genera or family was distributed in distinct week is low. This low possibility reduces the individual influence of families and genus to community turnover patterns.

4.1. Community OTUs turnover aspects in succession

The most remarkable change in community dynamics regarding time of succession is the decrease in the accumulation of new taxa, as predicted by the first hypothesis and corroborated by the results. This pattern was observed in other communities, both terrestrial and aquatic (Anderson, 2007; DeVries et al., 1999; Strayer et al., 2006). However, the intensity of decline observed for aquatic insects in temporary streams was high. Taxa accumulation decreased from 6 to 1.6 taxa after one week. The organisms inhabiting temporary habitats need two different strategies to deal with environmental variability, namely drought resistant forms (like dehydrated eggs and cocoons) or pronounced dispersal (Southwood, 1977). Abundant water availability is essential to most aquatic insects, especially in their immature stages (Merritt et al., 2008). Therefore, the main process sustaining aquatic insect communities in temporary streams would be a combination of short life cycles and rapid dispersal and colonization.

Communities living in ephemeral habitats are almost entirely composed of opportunistic species, which take advantage of the new habitat available, since there are a restricted number of taxa specialized on temporary habitats (Kavazos and Wallman, 2012). The progressive increase of richness on a temporal scale is a recurrent pattern observed in temporary streams (Bogan and Boersma, 2012; Carvalho and Uieda, 2006). However, the experiment showed a constant turnover of taxa in the streams instead of taxon accumulation. We recorded a high rate of taxon accumulation at the beginning of the experiment, followed by a rapid decrease of that rate, and a subsequent increase in community turnover (test results for the second to fourth hypothesis). This pattern indicates that the community of aquatic insects is possibly mostly composed of highly dispersive and adaptable. Habitat colonization by aquatic insects in the early stream overflow is likely a key process in the dynamics of the ecosystem of temporary streams.

Aquatic insects are important biodiversity components in aquatic environments, carrying out a variety ecosystem functions (e.g., nutrient cycling, energy transfer, secondary production, control of lower and upper trophic levels, etc.; Baptista et al., 2007; Resh, 2008), and are important biodiversity components. Interestingly, aquatic insects generally have a great phenotypic plasticity and, for this reason, their functional roles are primarily related to their trophic mode instead of their taxonomy (Cummins et al., 2005). Other relevant groups, e.g., fish and algae, have their population dynamics linked with aquatic insects communities by biotic interactions like competition or herbivory (Merritt et al., 2008). Therefore, rapid insect colonization enables a rapid stabilization of lotic ecosystem functions in temporary streams, mainly as regards to energy transfer. Although the number of new colonizers at the beginning of the experiment was high, there was an increase in the number of new taxa over time, regardless of the reduction on increment rates. The lack of difference between artificial habitats originally placed in the stream and those that replaced them indicate that the age of the community (new or well-established), had little influence on identity of new colonizers.

4.2. Metacommunity importance to successional dynamics

Previous studies had already observed temporal changes in

species richness, total abundance of individuals, and relative abundance of species in intermittent streams (Boulton and Lake, 1992). In tropical permanent streams the main insect colonization route is downstream, with individuals moving actively or passively by drift in the longitudinal axis of the stream and crawling through the other two dimensions (transversal and water column) following the substrate (Carvalho and Uieda, 2006). Although downstream movement is important, the aerial route is one of the most important sources of aquatic insects in the early colonization of temporary streams (Bogan and Boersma, 2012; Landeiro et al., 2011; Williams and Hynes, 1976). The aerial route depends on the source of adult insects in the region, and the oviposition in a specific stream is an arbitrary event, leading to the elevated community turnover rate observed each week. This reasoning is enhanced by the great difference among recently settled and previously established communities, reflecting a shift in which taxa colonize the substrate each week.

The difference in community turnover among weeks shows the importance of the landscape for the community succession in some streams. Streams with high weekly community dissimilarity were the ones closest to the Maranhão River, a permanent watercourse. Such proximity may have favored the arrival of the colonizing individuals from the river to the stream, and beget unpredictable community dynamics, with pronounced community turnover. The differential colonization resulting from landscape configuration (i.e., the proximity to the species source; intermittent river) defines the dependence on dispersal and colonization of aquatic insects on spatial structure. The difference in taxa turnover observed in the streams (five hypothesis test) allow us to address the importance of landscape configuration for the successional dynamics in the metacommunity studied.

In the metacommunity concept, aquatic insect communities of temporary streams have elements of succession similar to those of insect larvae communities in fruits rotting on forest ground (Atkinson and Shorrocks, 1984). These communities are strictly dependent on ephemeral and seasonal environmental conditions, and diversity maintenance on a local scale is an emergent feature that results from the colonization capacity of individual species. Therefore, each temporary stream has a singular colonization history, product of the relationship between the spatial structure of the river basin and the dispersal traits of the species.

4.3. Biological functional patterns

Community variation over time caused a change in the functional structure of the aquatic insect community beyond the composition turnover, primarily by the substitution of different functional groups. The initial community was composed basically of Diptera (flies) and Ephemeroptera (mayflies), two groups with high colonization rates (Carvalho and Uieda, 2006; Murrell et al., 2014) and relatively fast development (Merritt et al., 2008). The flies and mayflies that initially occurred are included in the collector-gatherer and scraper feeding groups (Oliveira and Nessimian, 2010). Organisms of these feeding groups process medium to large fragments of organic matter or remove lightly attached/loosely deposited organic material. Odonata was the main predator at the beginning of the experiment, and its abundance decayed as time advanced. Dragonflies have a high dispersal capacity (Mendes et al., 2015), which may enable quick colonization of new habitats.

The turnover in functional traits within a community is a consequence of the effects of different requirements and capacities on dispersal of the taxa involved in community succession in temporary streams. Such dependence may be observed in the chronological colonization events observed for the insect orders.

The number of Trichoptera genera began to rise by the middle of the succession sequence. Meanwhile, in the predator group, Heteroptera and Megaloptera replaced the Odonata. At the end of experiment, the community was basically composed of Trichoptera. The Trichoptera order is a diverse functional group, comprising predators, herbivores, collectors and shredders. The high proportion of Trichoptera at late successional stages may be due to their efficient use of the environment (De Moor and Ivanov, 2008). However, this would need to be tested by specific experiments. The late successional taxa, mainly Heteroptera, have a relatively limited dispersal capacity compared to the initial taxa, given that their main route of movement is through the stream or that the species need environment with more species and individuals to inhabit, owing to their predatory habits.

5. Conclusion

The importance of landscape for the colonization and succession pattern of the community of aquatic insects in intermittent streams now becomes apparent with this study. The possibility of a rapid colonization of new habitats in temporary streams indicates the connectivity of different communities with a large interchange of individuals. We noted that overland dispersal is likely an important component in the colonization of temporary streams. The successional processes in communities inhabiting ephemeral habitats have similar elements, such as the rate of accumulation of new taxa and the turnover of functional traits in the community at different times. However the spatial component may be relevant in the successional dynamics, since different streams would have different successional history.

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